



Concentration has broken complex systems before. AI now brings it to judgment.

Every case below runs the same mechanism: as diversity flattens and people come to think alike, their errors stop cancelling out and strike together instead of averaging away.

LTCM, 1998
\$4.6 bn

A fund run by Nobel laureates converged on identical arbitrage strategies, so its many positions were really one bet. When it turned, \$4.6 bn vanished in four months, nearly taking Wall Street down too.

Banks, 2008
all at once

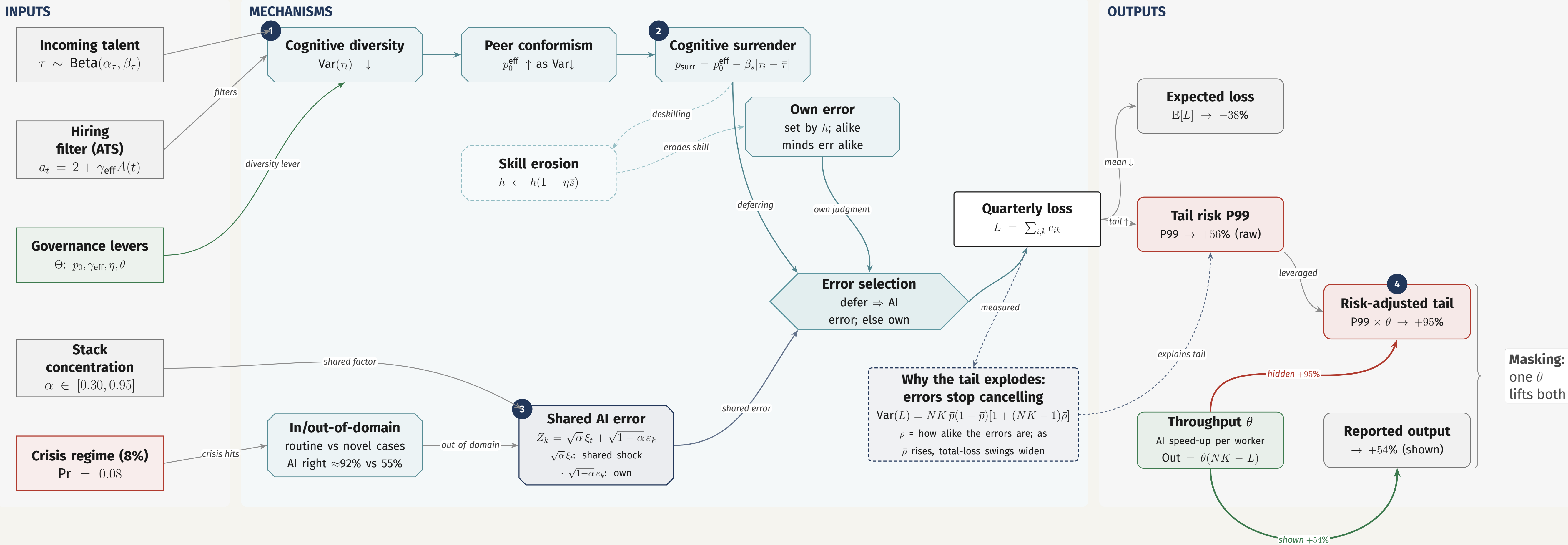
Banks worldwide priced risk with the same value-at-risk models and the same Gaussian copula. One shared formula, so they broke everywhere at once, not bank by bank.

CrowdStrike, 2024
8.5 M down

One security update shipped to Windows hosts everywhere. In 78 minutes it crashed 8.5 million machines and cost the Fortune 500 \$5.4 bn. One vendor, one dependency, with no fallback once it failed.

Shared AI lowers the average error, then concentrates the risk

AI now drives that same convergence inside the firm, on human judgment itself. Three vendors already hold **88%** of enterprise AI spend.



- 1 The filter narrows minds.** One algorithm screens every CV with one logic, so atypical profiles drop out, and internal promotion keeps compounding the sameness.¹ In the model, expressed cognitive diversity falls **–29%** over five years.
- 2 Surrender becomes the default.** Under productivity pressure, deferring is efficient and spreads as a habit: time pressure triples blind surrender, and even when the AI is wrong **79.8%** follow it.²
- 3 One shared error.** *In-domain*: routine cases inside the AI's reliable frontier, where it is right ~92%. A crisis (8% of quarters) pushes work *out-of-domain*, beyond that frontier (~55%),³ so everyone errs together.
- 4 The masking dashboard.** Errors now line up, so losses stop averaging out and pile into a few extreme quarters. A single speed-up lifts the output you see (+54%) while hiding that risk (P99×θ +95%).

¹Rivera 2012 (hiring & promotion homophily) · ²Shaw & Nave 2026 (deference to AI) · ³Dell'Acqua 2026 (the jagged frontier) · shared-error factor (schema): Vasicek 2002.

The risk the dashboard cannot see

No internal fix is enough, and the window is closing

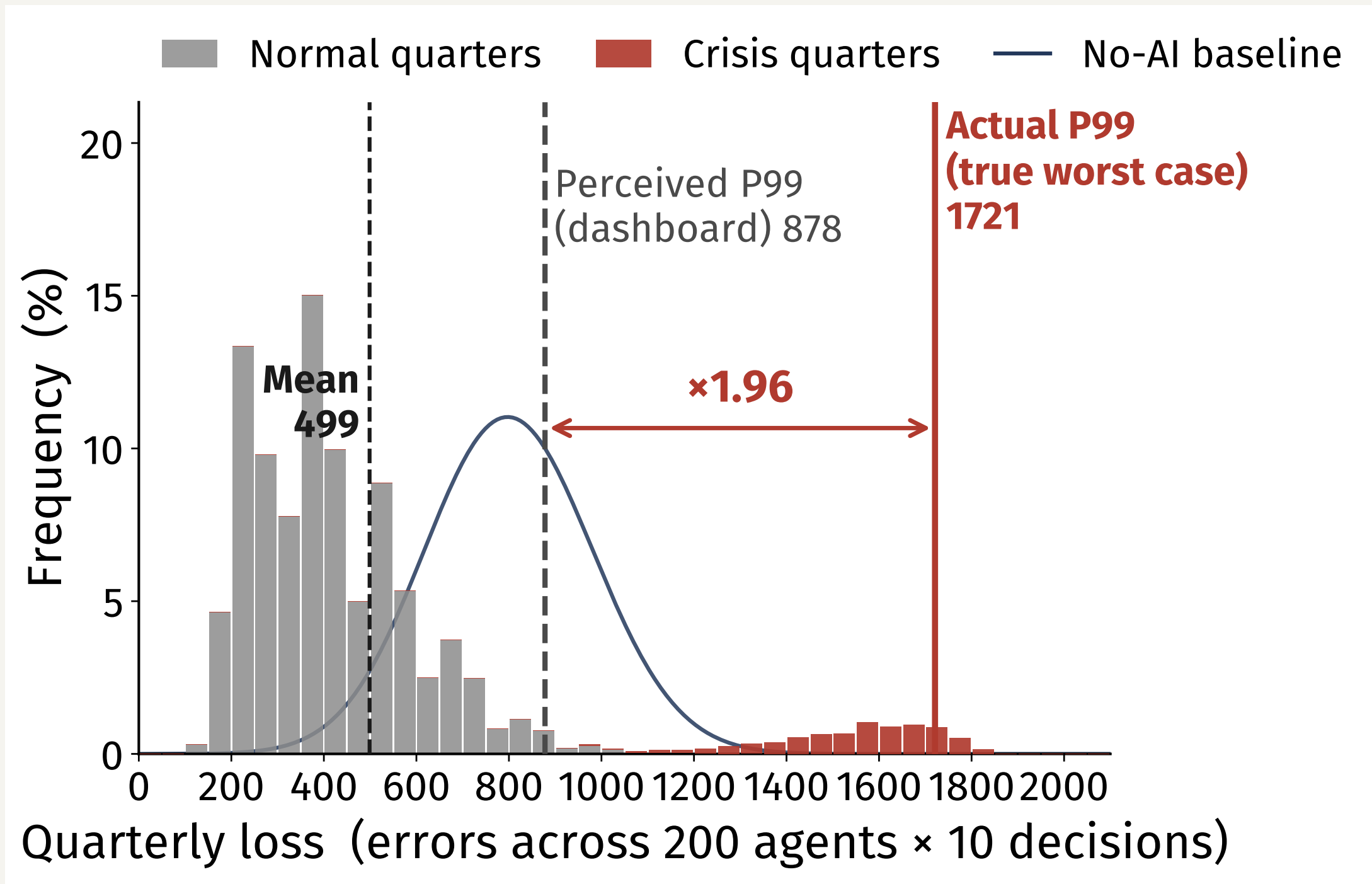


Fig. 1 · The tail is always underestimated. A dashboard that sees only normal quarters calls 878 the worst case; the true tail (1721) is 1.96× as large, unseen.

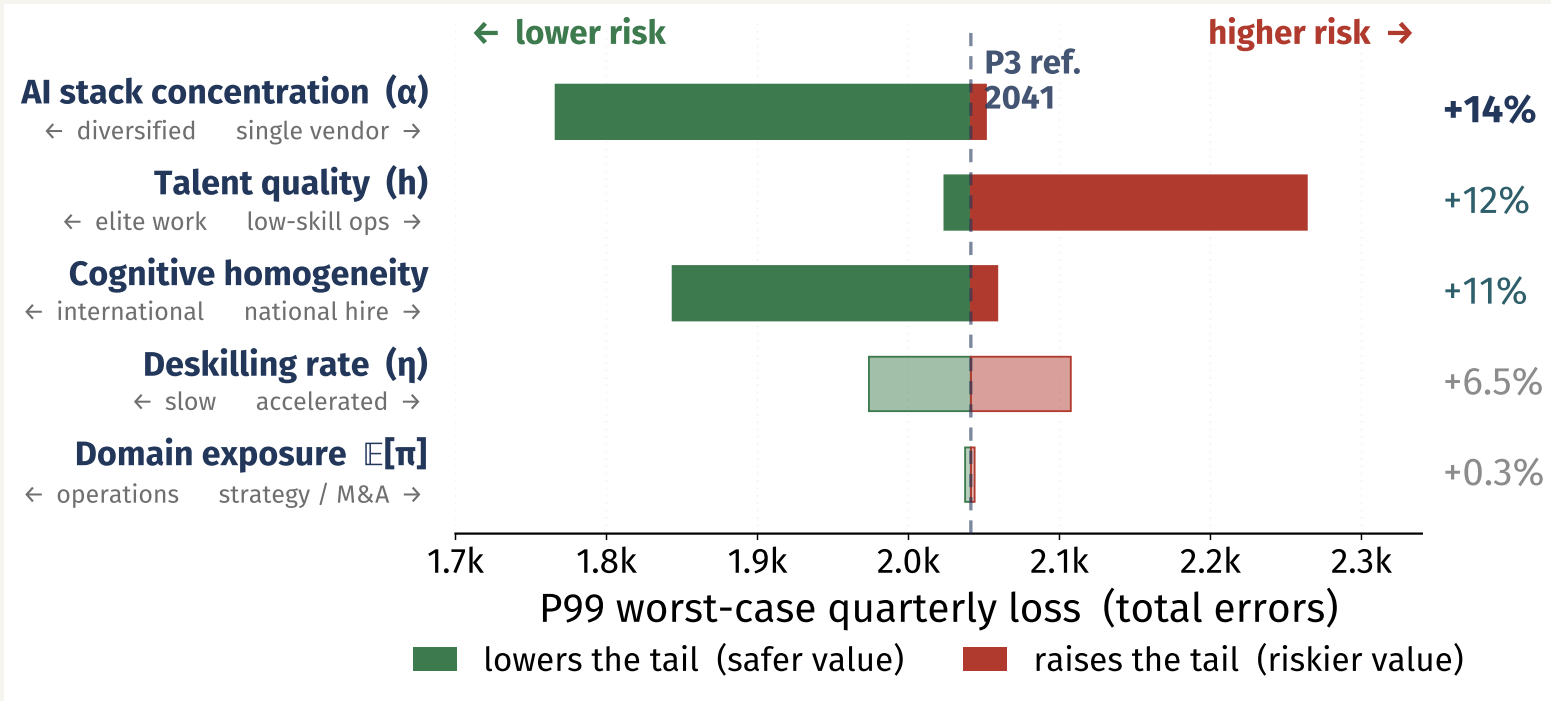


Fig. 3 · What moves the tail (left). Swept one at a time, three drivers stand out: stack concentration (α), talent quality (h: elite firms lose more) and cognitive homogeneity, each adding 11 to 14%; the rest barely move it. The channels feed back on each other, so their effects do not simply add up.

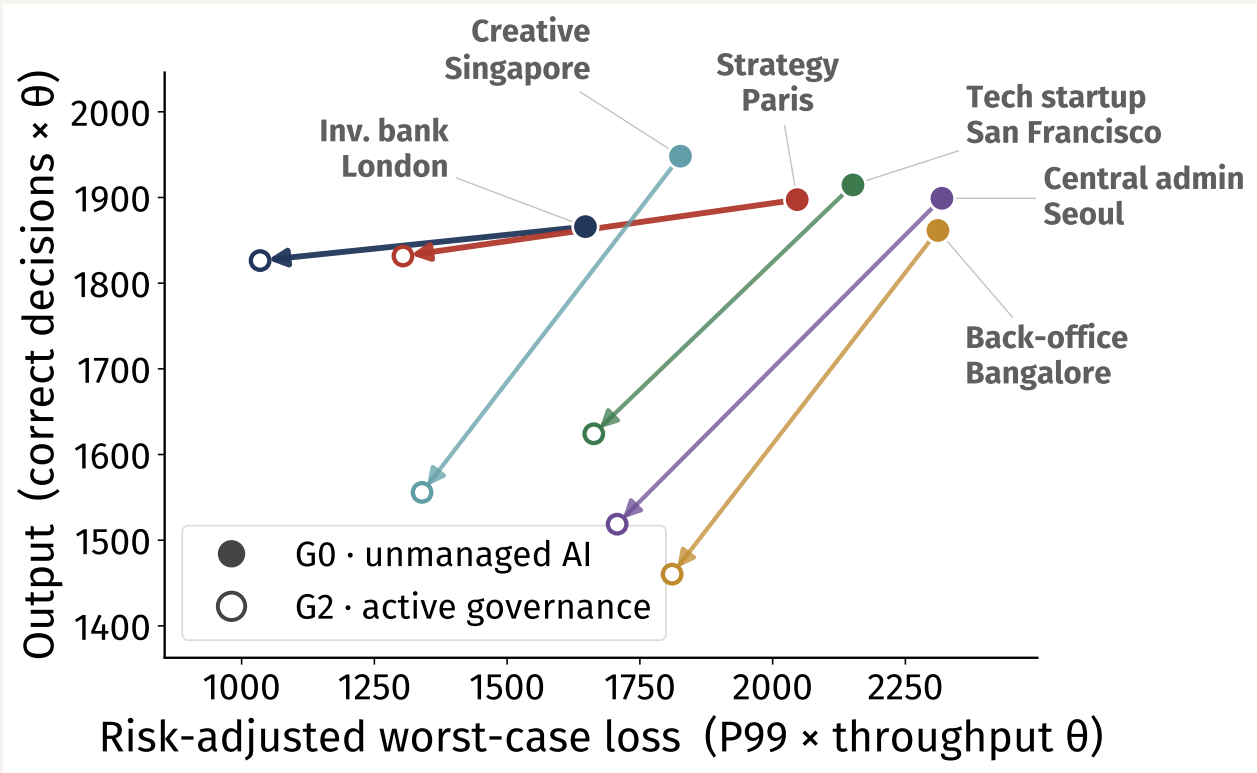


Fig. 4 · Governance helps, unevenly (right). Active oversight (G2) lowers every profile's risk, but *regressively*: it helps least the firms most exposed, whose elite judgment is exactly what AI erodes fastest.

G0 no oversight · G1 passive scaffold (the AI flags its weak answers) · G2 active scaffold = independent judgment formed before the AI is consulted.

Mitigation levers: owner and effect

Lever	Owner	Action and effect
Stack diversification	Procurement	Different model at each layer. Tail –10 to –15%.
Active scaffolding	COO / BU	Independent judgment before the AI. Excess tail risk +95% → +39%.
Hiring governance	CHRO	Audit the hiring AI for sameness; protects cognitive diversity.
Risk monitoring	CRO / Board	Track error-correlation; report P99×θ, not output.

Why the problem will not self-correct.

- ▶ **Small firms do not carry the tail.** A start-up that fails walks away from its losses, leaving the large incumbents to absorb them; running AI ungoverned (a +54% output edge) is rational, so competition pulls everyone in.
- ▶ **Deskilling does not reverse.** When AI was withdrawn from endoscopists trained on it, their detection rate fell below the pre-AI baseline; a workforce that has not judged independently for years cannot resume it on demand, and the fix costs more each quarter.
- ▶ **No backstop is coming.** The EU AI Act (August 2026) sets no concentration limit, and ~100,000 mid-market firms (big enough to deploy AI, too small to govern it) sit in the path of the three dominant vendors.

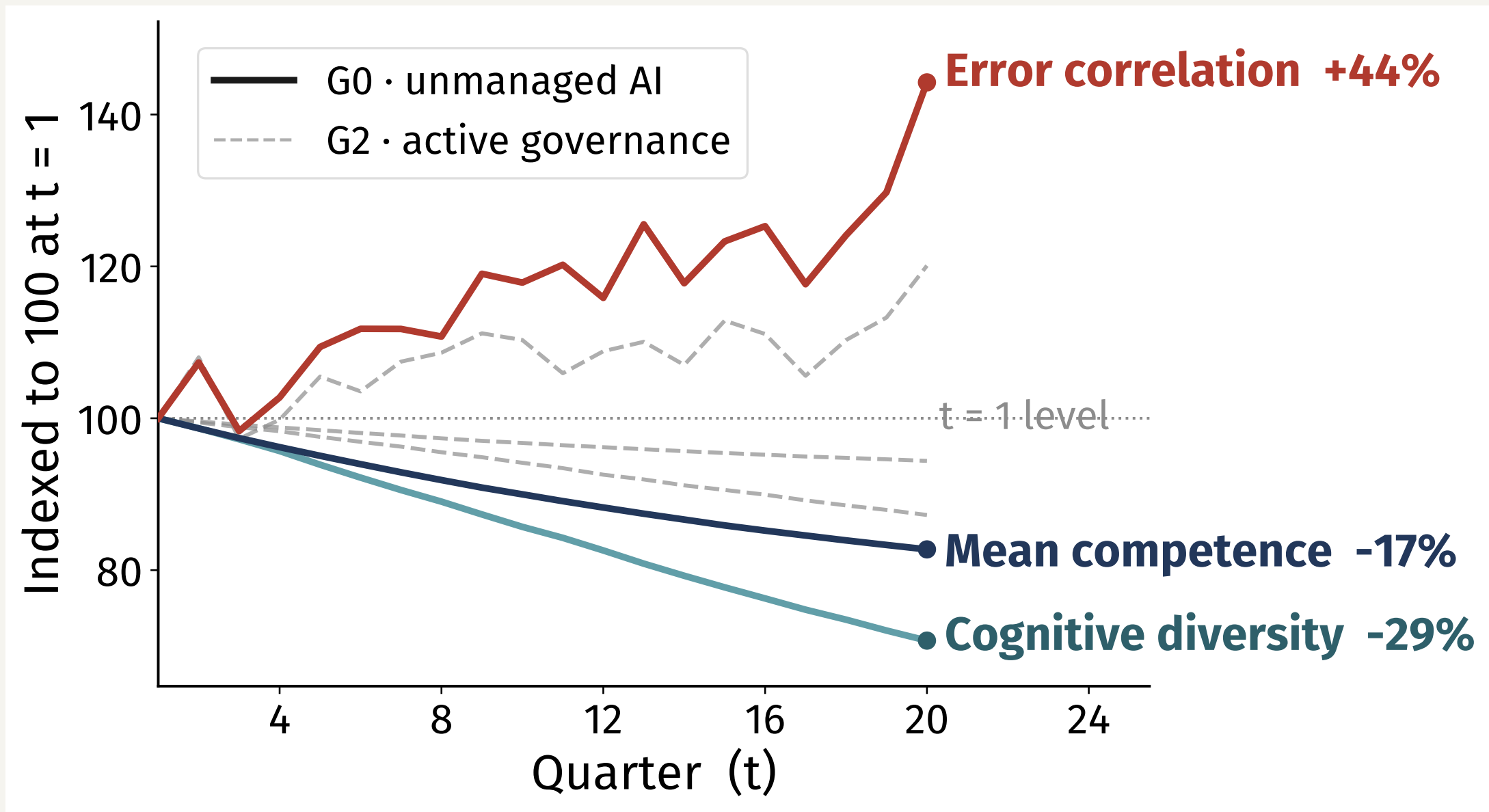


Fig. 2 · The damage compounds for years. Left unmanaged, people think alike (diversity –29%), skills fade (–17%), and errors correlate (+44%). After five years the loss is built in; active governance (G2) holds it flat.